DATABRICKS

FINOPS

Cost Optimization & Financial Governance on Azure

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Azure Well-Architected Framework

Quality-driven tenets, architectural decision points & review tools to build solid Azure workloads

What is the WAF?

A set of quality-driven tenets, architectural decision points, and review tools to help architects build a strong technical foundation for Azure workloads.

Supported Workloads

- AI / Generative AI
- SaaS (ISV & Startups)
- SAP on Azure
- Mission-Critical
- Analytics
- Azure VMware Solution
- Azure Virtual Desktop
- Sustainability

The 5 Pillars

Reliability

Redundancy, resiliency, uptime & recovery targets at scale

Security

Confidentiality, data integrity, protection against attacks

Cost Optimization

Spend within budget at organizational, architectural & tactical levels

Operational Excellence

Holistic observability, automated systems, fewer production incidents

Performance Efficiency

Horizontal scaling, test changes before deployment,

How to Use It?

1. Choose the Pillars most relevant to your workload
2. Align your architecture decisions with the principles
3. Build your workload foundation using design areas
4. Assess your readiness with the Assessment tool

Azure Architecture Review

Free assessment tool to measure your production readiness across all 5 pillars.

AGENDA

1 **Azure Databricks Pricing Model**

DBU, clusters, storage & consumption model

2 **Cost Monitoring**

System tables, Azure Cost Management

3 **Tagging for Cost Attribution**

Unity Catalog, cluster tags, propagation

4 **Discount Strategies**

PAYG, Commit Units, Savings Plans, Reserved VMs

5 **Data & Cluster Optimization**

Serverless, Cobalt 100 VMs, best practices

Azure Databricks

Pricing Model

Cost = # DBU × Rate ($) × Usage Time

3 main components drive your bill:

- **DBU Consumption**
- **Cluster Types & Config**
- **Storage Costs**

What is a DBU?

A Databricks Unit measures processing capability per hour — the basis for all platform billing.

- Job scheduling
- Cluster management
- Notebooks
- Data processing

⚠ Cloud compute (VMs, storage, networking) is billed separately by Azure

Monitor with System Tables

Databricks-hosted analytical store in the system catalog for historical observability across your account.

Table	Description	Retention	Data Scope
billable_usage	Billable usage records across your account — cost per cluster	1 year	Global
clusters	Cluster lifecycle events (start, terminate, resize)	1 year	Regional
jobs	Job runs, task-level execution details and duration	1 year	Regional
node_types	Available VM types and their DBU rates	Current	Regional
warehouses	SQL warehouse usage and query metrics	1 year	Regional

💡 Query system.billing.billable_usage for complete cost attribution at cluster level

Monitor with System Tables

The SQL editor now supports sessions

You can now define variables or create temporary views in one statement and reuse them in later runs.

Run all (1000) | 06:21 PM (2s) | ric44 . default | New SQL editor: ON | Last edit was 3 minutes ago

```sql
SHOW GRANTS ON CATALOG system;
SHOW SCHEMAS IN system;
SHOW TABLES IN system.billing;

SELECT * FROM system.billing.usage LIMIT 10;

SELECT * FROM system.compute.warehouses LIMIT 10;

SELECT * FROM system.access.audit LIMIT 10;


SELECT
    sku_name,
    usage_unit,
    SUM(usage_quantity) AS total_usage
FROM system.billing.usage
GROUP BY sku_name, usage_unit
ORDER BY total_usage DESC;

```

Add parameter

Table

	sku_name	usage_unit	total_usage
1	PREMIUM_ALL_PURPOSE_COMPUTE	DBU	73.356226111111111111
2	PREMIUM_ALL_PURPOSE_SERVERLESS_COMPUTE_US_WEST_3	DBU	0.886969722222222254
3	PUBLIC_CONNECTIVITY_DATA_PROCESSED	GB	0.000182777000000000
4	INTERNET_EGRESS_NORTH_AMERICA	GB	0.000048081000000000

Demo

Azure Cost Management

Cost Analysis

View and analyze spending by resource group, resource, tag, or time period. Filter by Databricks workspace to isolate costs.

Budgets & Alerts

Set spending thresholds and receive alerts before budget overruns. Configure by subscription, resource group or tag.

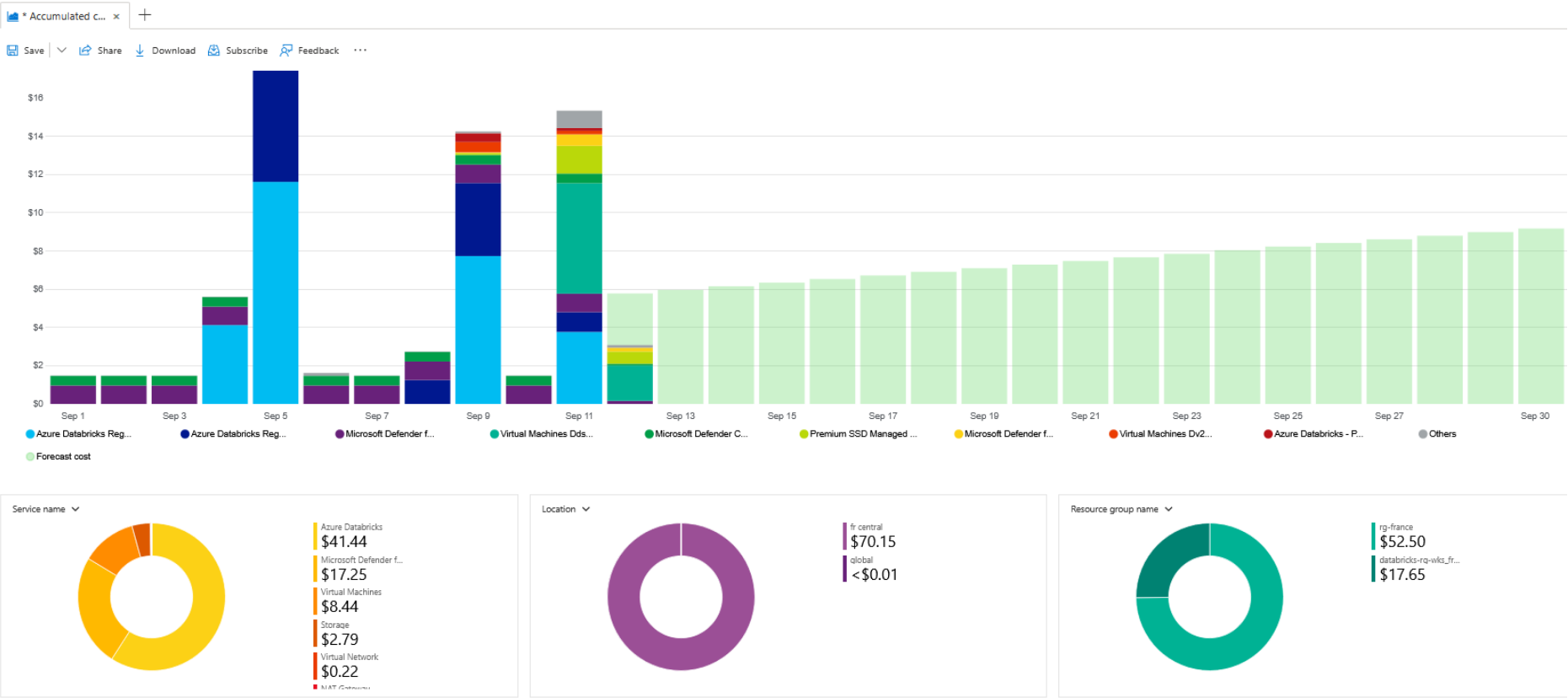
Tag-Based Attribution

Use cluster tags propagated to Azure VMs for granular cost attribution per team, project or environment.

Cost Exports

Export usage data to storage accounts for integration with Power BI, custom dashboards, or external FinOps tools.

Azure Cost Management



Tagging for Cost Attribution

Unity Catalog Object Tags

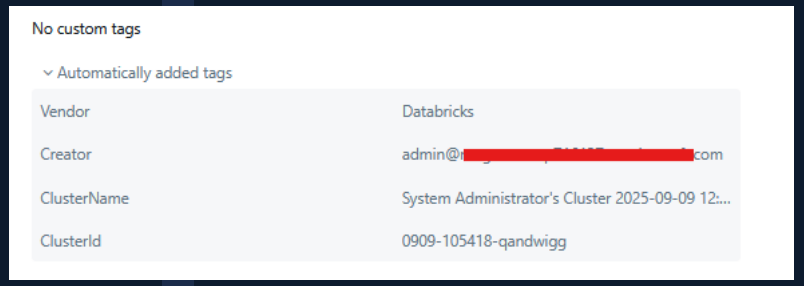
- Tag Column, Table, Schema, Catalog objects
- Search and filter objects by tag
- Governance and lineage tracking

Cluster Tags → Cloud Provider

- Tags propagated to Azure VMs
- Visible in Azure Cost Management
- Used for billing attribution

Default Databricks Tags

- Auto-applied by Databricks to resources
- Compute: vendor, clusterID, clusterName
- Lakehouse Monitoring: lakehouseMonitoring



💡 Consistent tagging across Unity Catalog + cluster config = full cost transparency

Tagging — SQL Examples

Tag a Table

```
ALTER TABLE sales.orders
SET TAGS ('sensitivity' =
'confidential');
```

Tag a Column / Multiple Tags

```
ALTER TABLE sales.orders
ALTER COLUMN customer_email
SET TAGS ('pii' = 'true’);


ALTER TABLE sales.orders
SET TAGS (
  'domain' = 'finance',
  'owner' = 'finance_team',
  'retention' = '7y'
);
```

Finops Tags

```
-- Cost allocation by domain (Data
Mesh / Business Unit)
ALTER TABLE sales.orders
SET TAGS ('domain' = 'finance’);

-- High cost data
ALTER TABLE raw.clickstream
SET TAGS ('criticality' = 'low’);

--Reduce data sprawl across the
organization
No owner
No retention
```

Serverless Budget Policies (Public Preview)

Use serverless budget policies to attribute serverless compute usage. Note: serverless compute usage is logged in billable usage system tables.

Tag Propagation Flow

Workspace Tags
Environment, team, project

→

Pool Tags
Pool-level attribution

→

Cluster Tags
Aggregated & applied

→

Azure VMs
Visible in Cost Management

Propagation Rules

- Tags from workspace, pools, and clusters are aggregated by Azure Databricks
- Applied to Azure VMs for cost reporting in Azure Cost Management
- Workspace + Pool tags ⟹ propagated to cluster resource → propagated to VMs
- Limit: max 50 tags per resource (Azure limit)

Pricing & Discount Strategies

1

Pay-as-you-go

Billed by compute capacity per second. No upfront commitment or minimum usage. Maximum flexibility.

Baseline

2

Commit Units (DBCU)

Prepurchase DBUs for 1 or 3-year terms. Units don't expire hourly. Applies to all Databricks workloads and tiers.

DBU Discount

3

Azure Savings Plan

Fixed hourly commit (e.g. $5/hr) for 1-3 years on compute. Up to 65% discount. Covers Databricks & other Azure compute.

Up to 65% off

4

Reserved VM Instances

1 or 3-year VM reservations for underlying Databricks nodes. Ideal for stable, predictable workloads. Supports family flexibility.

Predictable

5

Combine RI + Savings Plan

Permanent clusters → Reserved Instances. Variable/job clusters → Savings Plan. Maximize coverage without overlap.

Best of both

Azure Savings Plan — Example

Scenario: 4-worker cluster, VM Standard_D4lds_v5

PAYG Rate:	$0.424/hr per VM
Savings Plan Rate:	$0.25/hr per VM (41% discount)
Hourly Commitment:	$1.20/hr

Hour 1

Cluster running

PAYG	Savings Plan
4 × $0.424 = $1.696/hr	$1.20 (commit)

$0.496 saved

Hour 2

Cluster off (idle)

PAYG	Savings Plan
$0/hr	$1.20 (still charged)

–$1.20 (wasted)

💡 Savings Plans reward consistent high-utilization workloads. Use for always-on clusters.

Reserved VM Instances

Commitment Terms

1-year or 3-year term commitments for Azure VMs underlying Databricks driver and worker nodes.

Family Flexibility

Use family flexibility to resize VMs within the same family (e.g. Standard_E8s_v3 → Standard_E16s_v3).

Eligible VMs

Standard_DS3_v2, Standard_E8s_v3 and other VMs used as driver and worker nodes.

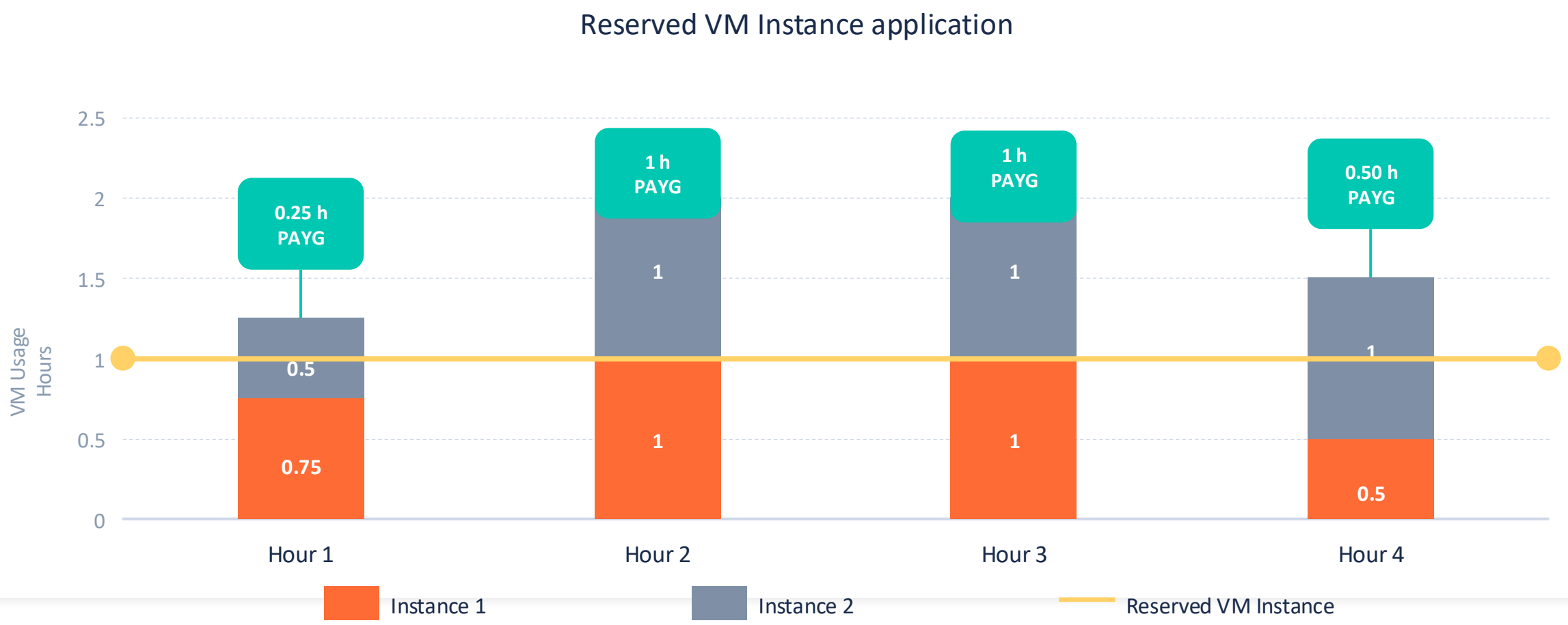
⚠ Important Behaviors

- Use-it-or-lose-it: If you don't have matching usage, reservation hours are wasted.
- Deallocate or delete VM → reservation discount automatically applies to another matching resource in scope.
- Stopped VMs are still billed and continue to consume reservation hours.

Azure Databricks Pricing Model

Infrastructure cost strategies to reduce your Azure bill

Reserved VM Instances



Optimization Strategies

Databricks Serverless

- ✓ Infrastructure costs included
- ✓ No idle clusters → auto-scale to zero
- ✓ Often more cost-effective for short jobs
- ✗ Higher DBU rate vs. dedicated clusters
- ✓ Logged in billable_usage system table
- ✓ Budget policies for attribution

Best for: short-running jobs, bursty workloads, and development environments

Cobalt 100 VMs

40–50%

better price/performance vs. previous gen

Dpsv6-series	General purpose
Dplsv6-series	General purpose
Epsv6-series	Memory-optimized

Standard compute category → eligible for Savings Plan discounts when not covered by RI

Data Optimization & Cluster Management

Autoscaling Clusters

Enable autoscaling to match actual workload needs. Avoid over-provisioning by setting appropriate min/max workers.

Cluster Lifecycle Policies

Set auto-termination (e.g. 10-30 min idle). Use cluster pools to reduce spin-up time while controlling costs.

Delta Lake Optimization

Use OPTIMIZE + ZORDER for faster queries. VACUUM to remove old files. Auto Optimize and Auto Compaction reduce small file overhead.

Query & Job Optimization

Profile slow jobs. Use photon engine where cost-effective. Partition pruning, broadcast joins, and caching reduce compute time.

Right-Sizing

Match VM family to workload: memory-optimized for ML, compute-optimized for ETL. Use spot/spot instances for fault-tolerant jobs.

Storage Optimization

Use managed Unity Catalog storage with lifecycle policies. Delta sharing avoids data duplication. DBFS external locations for cost control.

Databricks FinOps

Practical tips to reduce your Azure Databricks bill

5 Tips to Save Money on Databricks

01 Enable Autoscaling & Auto-termination

Configure clusters to adapt to actual load and shut down after 10–20 min of inactivity. An idle cluster running overnight can waste hundreds of dollars per month.

Quick win

02 Use Serverless for Short Jobs

Despite a higher DBU rate, serverless eliminates infra costs and startup time. For jobs under 30 min or irregular workloads, it's often cheaper than a dedicated cluster.

Cost-effective

03 Optimize Your Delta Lake Tables

Run OPTIMIZE + ZORDER on frequently queried tables and VACUUM to remove stale files. Fragmented small files multiply read operations — and your compute bill.

Performance

04 Combine Reserved Instances + Savings Plan

Cover predictable clusters with Reserved VM Instances (up to ~60% off). Use Azure Savings Plan for variable/job clusters. Don't try to cover everything with one mechanism.

Up to 60% off

05 Tag Everything from Day One

Without consistent tags on clusters, tables and workspaces, you can't see who spends what. Define a tagging policy (team, project, env) in Unity Catalog to surface and eliminate hidden costs fast.

Governance

Thank You

Questions & Discussion

System Tables for observability

Tagging for cost attribution

RI + Savings Plan = max discount

Serverless + Cobalt for efficiency